

Curriculum Vitae of
Md Mutasim Billah, Ph.D.

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Academic Appointment

- **Full-time Lecturer** Fall 2026–Present
 - Department of Mathematics, Applied Mathematics, and Statistics (MAMS)
 - **Case Western Reserve University**, Cleveland, Ohio
- **Visiting Assistant Professor** Fall 2025–Summer 2026
 - Department of Mathematics, Statistics, and Computer Science (MSCS)
 - **Macalester College**, St. Paul, Minnesota

Education

- **Ph.D.** in Statistics Fall 2023–Summer 2025
 - **Michigan Technological University**, Houghton, Michigan
 - **Dissertation**: “Methods in Statistics, Machine Learning, and Deep Learning for Combining Multi-Omics Datasets”
 - **Advisor**: Dr. Kui Zhang, Professor, Department of Mathematical Sciences
- **M.Sc.** in Statistics Fall 2019–Fall 2023
 - **Michigan Technological University**, Houghton, Michigan
- **Graduate Studies** in Applied Statistics Fall 2018–Summer 2019
 - **University of Northern Colorado**, Greeley, Colorado
- **M.Sc.** in Statistics, Biostatistics & Informatics Fall 2012–Summer 2013
 - **University of Dhaka**, Dhaka, Bangladesh
- **B.Sc.** in Statistics, Biostatistics & Informatics Fall 2007–Summer 2012
 - **University of Dhaka**, Dhaka, Bangladesh

Honors and Awards

- **Finishing Fellowship** from Michigan Technological University in Summer 2025
- **Recognition Award** for exceptional average of 7 dimensions in teaching in Fall 2020
 - Provost’s office for academic affairs, Michigan Technological University
 - Only 73 instructors (86 sections out of more than 1,000 evaluated) university-wide were rated this highly by students
- **Recognition Award** for outstanding teaching in Spring 2020
 - Provost’s Office for Academic Affairs, Michigan Technological University
- **Course Excellence Awards**, Department of Mathematical Sciences, Michigan Technological University
 - Awarded for top performance in graduate and Ph.D.-level coursework, including Probability I & II, Mathematical Statistics I & II, Computational Statistics, Predictive Modeling, Applied Generalized Linear Model, Categorical Data Analysis, and Teaching College Mathematics.

Teaching Experience

- **Instructor of Record**
 - Full-time Lecturer** Fall 2026–Present
Department of MAMS, Case Western Reserve University
 - Teaching undergraduate and graduate statistics courses while developing course materials, assessments, and active-learning activities to support student success
 - Visiting Assistant Professor** Fall 2025–Summer 2026
Department of MSCS, Macalester College

- Instructing courses at both the foundational and intermediate levels of the undergraduate statistics and data science curriculum

Graduate Teaching Instructor

Spring 2020–Spring 2025

Department of Mathematical Sciences, Michigan Technological University

- Instructed MA 3710: Engineering Statistics across multiple modalities: in-person, hybrid, synchronous online, and asynchronous online

Graduate Teaching Instructor

Fall 2018–Summer 2019

Department of Applied Statistics and Research Methods, University of Northern Colorado

- Instructed Introduction to statistics, a required general-education course

- **Teaching and Learning Support**

Graduate Teaching Assistant

Dr. Kui Zhang, Mathematical Sciences, Michigan Technological University

Fall 2022–Spring 2023

Beth Reed, Mathematical Sciences, Michigan Technological University

Fall 2019

- **Instructor of records/Co-instruction/Teaching Assistant course experience**

Duties, syllabi, and course materials are available on my [professional website](#).

Course	Institution	Semester
Basic Statistics for Engineering & Science	Case Western Reserve University	Fall 2026 [•] (3 sections)
Introduction to Statistical Modeling	Macalester College	Fall 2025 [•] (2 sections) Spring 2026 [•] (2 sections)
Statistical Machine Learning	Macalester College	Spring 2026 [•]
Engineering Statistics	Michigan Technological University	Spring 2025 [•] Summer 2024 [†] Spring 2024 [†] Fall 2023 [†] Summer 2023 [†] Spring 2023 [†] Summer 2022 [†] Spring 2022 [†] Fall 2021 [†] Summer 2021 [†] Spring 2021 [†] Fall 2020 ^{††} Summer 2020 ^{††} Spring 2020 [•] Fall 2019 ^{**}
Applied Generalized Linear Models	Michigan Technological University	Spring 2023 [*]
Design & Analysis of Experiments	Michigan Technological University	Fall 2022 [*]
Statistical Methods	Michigan Technological University	Fall 2019 ^{**}
Introduction to Statistical Analysis	University of Northern Colorado	Fall 2018 [•] Spring 2019 [•] Summer 2019 [•]

- Instructor of record (in person)

† Instructor of record (asynchronous online)

†† Instructor of record (synchronous online)

* Teaching assistant (asynchronous online)

** Teaching assistant (in person)

Student Mentoring

- **Capstone Project Mentor, MSCS, Macalester College** Fall 2025–Spring 2026
 - Max Clifford and Ryan Mickelborough. *Investigating Energy Grid Balancing Authorities and Renewable Energy Usage Using a Time Series Analysis Approach*. Fall 25 - Spring 26
 - Noah Shannon and Austin Mills. *What Drives Winning in College Basketball? – Evidence from the Mixed Effect Model*. Fall 25 - Spring 26

Research Agenda

- **Research Interests**
 - **Core Methodology:** Developing methods in statistics, machine learning (ML), and deep learning (DL) tailored to genomic data; Statistical frameworks for multi-omics integration; Methodological advances in missing data imputation and uncertainty modeling for high-dimensional, incomplete biological data; Causal inference and modeling of biological heterogeneity
 - **Application Domain:** Dissection of gene regulatory architecture across diverse tissues (pleiotropy, cis/trans-effects); High-resolution fine-mapping of causal variants to elucidate disease etiology; Translation of multi-omics discoveries into predictive biomarkers and public-health interventions; Genomic medicine and health-equity research in ancestrally diverse and historically underserved populations
 - **Computational Science:** Scalable and reproducible computational genomics for cohort-scale studies (GWAS, GTEx, dbGaP); Development of end-to-end analysis pipelines for secure, controlled-access omics data; High-performance computing implementations that couple DL with statistical genomics for large-scale discovery
 - **Pedagogical Scholarship and Innovation:** Evidence-based instructional practices, including flipped classroom models, active learning, and student-centered approaches to develop statistical reasoning and data literacy; Designing scalable alternative grading and equitable assessment frameworks for diverse class sizes; Integration of generative AI to enhance computational literacy and modern STEM curricula
- **Active and Future Research Plan**
 - Extending previously developed multi-tissue TWAS frameworks (e.g., TWAS-CTL, G-Boost-CTL) to leverage GWAS summary-level data
 - Benchmarking advanced ML and DL Models for cross-tissue gene expression imputation
 - Investigating the distinct roles and interactions of local and distant genetic variants in controlling gene expression across different tissues within the TWAS framework
 - Developing advanced ML and DL pipelines to impute missing values in multi-tissue gene expression datasets (e.g., GTEx reference panel)
 - Developing sparsity-inducing, power-adaptive procedures to identify and remove non-informative tissues in multi-tissue TWAS
 - Developing unsupervised clustering to characterize pleiotropic effects in GWAS

Research Experience

- **Doctoral Researcher** Fall 2022–Summer 2025
Department of Mathematical Sciences, Michigan Technological University
 - Developed TWAS-CTL, a computationally efficient multi-tissue learning framework that improves statistical power by adaptively weighting gene-expression predictors across tissues
 - Introduced G-Boost-CTL, an enhanced cross-tissue learner that incorporates GWAS-derived weighting to prioritize tissues carrying true genetic signal, increasing discovery power while preserving error control
 - Integrated nonlinear models (gradient-boosted trees, deep neural networks) into the TWAS pipeline, revealing complex regulatory effects missed by standard linear approaches and improving causal gene identification
- **Graduate Research Assistant** Summers 2023, 2024
Dr. Kui Zhang, Professor, Department of Mathematical Sciences, Michigan Technological University
 - Secured NIH dbGaP approval and coordinated the download of > 2 TB of protected GTEx genotype and RNA-seq data; built automated Bash/R pipelines for de-identification, quality control, and harmonization across 49 tissues
 - Designed and coded scalable R-/Python-based workflows for both established (PrediXcan, UTMOST)

and new cross-tissue TWAS methods (TWAS-CTL, G-Boost-CTL)

- Conducted large-scale power simulations, statistical validation, and data-visualization that underpin three forthcoming manuscripts

- **Statistical Consultant, Research Consulting Lab**

Fall 2018–Summer 2019

Department of Applied Statistics and Research Methods, University of Northern Colorado

- Acted as a lead statistical consultant, overseeing client intake, aligning project needs with consultant expertise, and mentoring new team members while coordinating statistical training workshops
- Delivered full-spectrum consulting support for qualitative, quantitative, mixed-methods, and evaluation studies – from research design and methodology development to final data analysis and reporting
- Designed and executed rigorous statistical analysis plans tailored to diverse academic disciplines
- Collaborated with graduate students, faculty, and staff on data analysis, research presentations, and thesis projects using R, SAS, and SPSS

Publications

- **Manuscripts Under Review**

- **Billah, M.**, Wei, H.[†], Sun, F.[•], Zhang, K.[†] (2025). *TWAS-CTL: A robust and efficient framework in multi-tissue transcriptome-wide association studies using cross-tissue learner*.
 - Submitted to *PLOS Genetics*; Preprint is available on [bioRxiv](#)
- **Billah, M.**, Wei, H.[†], Sun, F.[•], Zhang, K.[†] (2025). *GBoost-CTL: A novel method in multi-tissue transcriptome-wide association studies in cross-tissue learner incorporating GWAS information*.
 - Submitted to *PLOS Genetics*; Preprint is available on [bioRxiv](#)

- **Manuscripts in Preparation**

- **Billah, M.**, Acheampong, S., Kareem, K., Subedi, M., Silwal, S., Zhang, K.[†]. *Application of Deep Learning and Boosted Tree Methods in multi-tissue TWAS framework*.
 - Target journal: *PLOS ONE* (expected submission: Summer 2026)
- Kareem, K., Subedi, M., **Billah, M.**, Acheampong, S., Silwal, S. *Unsupervised Clustering of Pleiotropic Signals in GWAS Using Sparse Phenotype Networks*.
 - Target journal: *Scientific Reports* (expected submission: Summer 2026)

[†] Professor, Michigan Technological University • Professor, University of Southern California

Presentations

- **Seminars and Contributed Talks**

- **Billah, M.**, Wei, H., Sun, F., Zhang, K. *GBoost-CTL: A Novel Method for Multi-Tissue Transcriptome-Wide Association Studies Incorporating GWAS Information*. Research Seminar, Department of Mathematical Sciences, Spring 2025.
- **Billah, M.**, Wei, H., Sun, F., Zhang, K. *TWAS-CTL: A Robust and Efficient Framework for Cross-Tissue Transcriptome-Wide Association Studies*. Research Seminar, Department of Mathematical Sciences, Spring 2024.

- **Invited Talks**

- **Billah, M.**, Wei, H., Sun, F., Zhang, K. *TWAS-CTL: A Robust and Efficient Framework for Cross-Tissue Transcriptome-Wide Association Studies*. Rose Hulman Institute of Technology, Spring 2026.

- **Poster Presentations**

- **Billah, M.**, Wei, H., Sun, F., Zhang, K. *GBoost-CTL: A Novel Method for Multi-Tissue Transcriptome-Wide Association Studies Incorporating GWAS Information*. American Society of Human Genetics Annual Meeting, 2025.
- **Billah, M. (Presenter)** *A Simple Analogy for Research Plans in Grant Applications*. Prototype Poster Session, ACM Grant Workshop, 2025.
- **Billah, M. (Presenter)**, Acheampong, S., Kareem, K., Subedi, M., Silwal, S., Zhang, K. *Application of Deep Learning and Boosted Tree Methods in a Multi-Tissue TWAS Framework*. Graduate Research Poster Session, Michigan Technological University, Summer 2025.
- **Billah, M.**, Wei, H., Sun, F., Zhang, K. *TWAS-CTL: A Robust and Efficient Framework for Cross-Tissue Transcriptome-Wide Association Studies*. American Society of Human Genetics Annual Meeting, 2024.

- **Billah, M. (Presenter)** *Analyzing Dementia Using Open Access Series of Imaging Studies (OASIS) Data: A Comparison of Multiple Linear Regression and Beta Regression Approaches*. Research Evening, University of Northern Colorado, Spring 2019.

Grants

- **Grants in Preparation**
 - “Application of Deep Learning methods and its improved accuracy in predicting multi-tissue gene expression from genotypes”–R03 NIH Grant
- **Travel Grants**
 - Faculty Travel and Research Grant, Macalester College; Amount: \$1250 Fall 2025
 - Department of Mathematical Sciences Travel Grants, MTU; Amount: \$1000 Fall 2024
 - Graduate Student Government (GSG) Travel Grants, MTU; Amount: \$500 Fall 2024

Service and Leadership

- **Member, Transformation Committee, MSCS, Macalester College** Fall 2025–Spring 2026
 - Preceptor (TA) Program Development & Training
 - Faculty Support & Mentoring
 - DEI Initiatives
- **Treasurer and Vice President, Muslim Student Association** Spring 2021–Spring 2023
 - Graduate Student Government (GSG), Michigan Technological University
- **President, Bangladesh Student Association** Spring 2020–Spring 2021
 - GSG, Michigan Technological University

Professional Development

- **Workshops**
 - Associated Colleges of the Midwest Grant Workshop, Macalester College August 12-15, 2025
- **Conferences**
 - Upper Peninsula’s (UP) Teaching and Learning Conference, Bay College in Escanaba May 13-14, 2024
 - UP’s Teaching and Learning Conference, Lake Superior State University May 15-16, 2023
 - UP’s Teaching and Learning Conference, Northern Michigan University May 24-25, 2022

Computational, Instructional, and Analytical Proficiency

- **Statistical Programming and Data Analysis**
 - Proficient in **R**, **Python**, **SAS**, **PLINK**, **GCTA**, and **SPSS**
- **Instructional Design and Teaching Tools**
 - **Pedagogical Frameworks:**
 - Adept in inclusive frameworks including **Universal Design for Learning (UDL)** and **Equitable Assessment** (e.g., Alternative Grading, Specifications Grading).
 - Skilled in student-centered active learning methodologies such as **Project-Based Learning**, **Inquiry-Based Learning**, **Flipped Classroom** design, and the use of **Statistical Applets** to promote interactive learning, conceptual understanding, and data-driven reasoning.
 - **Computational Pedagogy:**
 - Specialized in teaching reproducible workflows by designing interactive course websites using **Quarto** and **R**, with version control managed via **GitHub**.
 - **Learning Technologies & Platforms:**
 - Proficient with Learning Management Systems (**Canvas**, **Moodle**) and a suite of instructional and accessibility technologies including **Panopto**, **Echo360**, **WebAssign**, **Gradescope**, **Speed-Grader**, **Respondus**, accessibility tools (**Grackle**, **UDOIT**), and audience response systems (e.g., **Poll Everywhere**, **iClicker**, **PointSolutions**).